UNIVERSITY OF THE PEOPLE

CS 2203-01 - AY2026-T4 Programming

Assignment Activity Unit 4

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Database Schema

This assignment implements a relational database schema for a university library management system. The schema is designed following the principles of structured data storage and relational integrity discussed in Unit 4. The database consists of three interrelated tables: Books, Members, and Loans. Each table stores specific entities with well,chosen SQL data types to ensure data integrity, minimize redundancy, and support efficient querying (Vidhya et al., 2016).

The three entities reflect a real,world library workflow: books are catalogued in the Books table, library members are registered in the Members table, and borrowing activity is tracked through the Loans table, which serves as a bridge linking Members to Books through foreign key relationships.

Part A: Database Schema Creation

Entity 1: Books Table

The Books table stores information about each book available in the library. The ISBN (International Standard Book Number) serves as the primary key, uniquely identifying each book. The data types are selected based on the nature of each attribute.

Attribute	Data Type	Constraint	Size	Justification

ISBN	VARCHAR	PRIMARY KEY, NOT NULL	20	ISBNs contain dashes; VARCHAR is flexible
Title	VARCHAR	NOT NULL	200	Variable,length; accommodates long titles
Author	VARCHAR	NOT NULL	150	Author names vary in length
Genre	VARCHAR	,	100	Categorical text; VARCHAR is appropriate
Quantity	INT	DEFAULT 0	,	Whole number; INT ensures no fractions

Entity 2: Members Table

The Members table stores registered library members. MemberID is an auto,incrementing integer primary key. Email and Phone are stored as VARCHAR to handle varying formats across different regions.

Attribute	Data Type	Constraint	Size	Justification
MemberID	INT	PRIMARY KEY, AUTO_INCREMENT	,	Unique numeric identifier, auto,generated
Name	VARCHAR	NOT NULL	150	Variable,length; suits diverse names
Email	VARCHAR	UNIQUE, NOT NULL	150	Stored as text; unique per member
Phone	VARCHAR	,	20	Can include +, dashes; VARCHAR fits

Entity 3: Loans Table

The Loans table tracks which member borrowed which book and the loan dates. It references both Books and Members tables via foreign keys, establishing referential integrity across the schema.

Attribute	Data Type	Constraint	Size	Justification
LoanID	INT	PRIMARY KEY, AUTO_INCREMENT	,	Auto,generated unique loan record ID
MemberID	INT	FOREIGN KEY	,	Links to Members table
ISBN	VARCHAR	FOREIGN KEY	20	Links to Books table
LoanDate	DATE	NOT NULL	,	DATE stores YYYY,MM,DD format
ReturnDate	DATE	,	,	NULL until book is returned

Part B: SQL Script with Comments

Step 1 , Create the Database and Tables (DDL)

The following SQL script uses Data Definition Language (DDL) commands to create the library database schema. The CREATE TABLE statements define the structure, constraints, and data types for each entity (Peterson, 2023).

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```sql
CREATE DATABASE IF NOT EXISTS LibraryDB;

USE LibraryDB;

CREATE TABLE Books (

    ISBN        VARCHAR(20)   PRIMARY KEY,      -- Unique book identifier
    Title       VARCHAR(200)  NOT NULL,         -- Book title
    Author      VARCHAR(150)  NOT NULL,         -- Author's full name
    Genre       VARCHAR(100),                   -- Genre category
    Quantity    INT           DEFAULT 0         -- Available copies

);


CREATE TABLE Members (

    MemberID    INT           PRIMARY KEY AUTO_INCREMENT, -- Auto ID
    Name        VARCHAR(150)  NOT NULL,         -- Member's full name
    Email       VARCHAR(150)  UNIQUE NOT NULL, -- Unique email address
    Phone       VARCHAR(20)                     -- Contact phone number

);

CREATE TABLE Loans (

    LoanID      INT           PRIMARY KEY AUTO_INCREMENT, -- Auto loan ID
    MemberID    INT           NOT NULL,         -- FK to Members table
    ISBN        VARCHAR(20)   NOT NULL,         -- FK to Books table
    LoanDate    DATE          NOT NULL,         -- Date book was borrowed
    ReturnDate  DATE,                           -- NULL until returned
    FOREIGN KEY (MemberID) REFERENCES Members(MemberID),
    FOREIGN KEY (ISBN)     REFERENCES Books(ISBN)

);

```

Step 2 , Insert Records (DML: INSERT)

The INSERT statements below populate all three tables with sample data. As noted in Vidhya et al. (2016), the INSERT command adds one or more rows to a table. Column,specific insertion is used here to improve clarity and maintain flexibility.

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```sql
INSERT INTO Books (ISBN, Title, Author, Genre, Quantity) VALUES
    ('978-0-13-468599-1', 'Database System Concepts', 'Silberschatz et al.', 'Computer Science', 5),
    ('978-0-07-352332-3', 'Operating System Concepts', 'Silberschatz & Galvin', 'Computer Science', 3),
    ('978-0-13-235088-4', 'Introduction to Algorithms', 'Cormen et al.', 'Mathematics', 4),
    ('978-0-321-12521-7', 'The Pragmatic Programmer', 'Hunt & Thomas', 'Software Engineering', 6);

-
INSERT INTO Members (Name, Email, Phone) VALUES
    ('Ahmed Hassan', 'ahmed.hassan@uopeople.edu', '+92-300-1234567'),
    ('Sara Malik', 'sara.malik@uopeople.edu', '+92-321-7654321'),
    ('John Carter', 'john.carter@uopeople.edu', '+1-555-0192');


INSERT INTO Loans (MemberID, ISBN, LoanDate, ReturnDate) VALUES
    (1, '978-0-13-468599-1', '2026-04-10', '2026-04-24'),
    (2, '978-0-07-352332-3', '2026-04-15', NULL),
    (1, '978-0-13-235088-4', '2026-04-20', NULL),
    (3, '978-0-321-12521-7', '2026-04-22', '2026-05-01');

```

Step 3 , Retrieve Records (DML: SELECT)

The SELECT query below retrieves all information about books borrowed by a specific member (Member ID = 1). A JOIN operation connects the Loans and Books tables through the shared ISBN column. The WHERE clause filters results to the specified member. This demonstrates the practical use of SELECT,FROM,WHERE,JOIN constructs described in Peterson (2023).

```sql
SELECT
    m.Name        AS Member_Name,
    b.Title       AS Book_Title,
    b.Author      AS Author,
    b.Genre       AS Genre,
    l.LoanDate    AS Loan_Date,
    l.ReturnDate  AS Return_Date
FROM Loans l
JOIN Members m ON l.MemberID = m.MemberID    -- Link Loans to Members
JOIN Books   b ON l.ISBN     = b.ISBN        -- Link Loans to Books
WHERE l.MemberID = 1;                        -- Filter for Ahmed Hassan

```

Step 4 , Update a Record (DML: UPDATE)

The UPDATE statement below modifies the quantity of a specific book in the Books table. As emphasized in the Unit 4 video by CBT Nuggets (2019), the WHERE clause is critical to ensure only the intended row is modified. Without WHERE, every record in the table would be affected.

```sql
UPDATE Books
SET    Quantity = 8
WHERE  ISBN = '978-0-13-468599-1';  -- Targets only this book

```

Step 5 , Delete a Member Record (DML: DELETE)

The DELETE statement removes a member from the Members table. A WHERE clause is mandatory to target only one record. The member being deleted (MemberID = 3, John Carter) must first have no active loan records, or the foreign key constraint would prevent deletion , demonstrating how referential integrity protects database consistency (Vidhya et al., 2016).

```sql
DELETE FROM Members
WHERE MemberID = 3;  -- Deletes only John Carter's record

```

Verification Queries

The following SELECT queries are used to verify that all operations executed correctly. These queries display all records in each table after the insert, update, and delete operations. Screenshots of these outputs should be captured from the SQL environment (e.g., MySQL Workbench or mycompiler.io) and submitted with this document.

```sql
-- Display all records in Books table (should show updated Quantity)
SELECT * FROM Books;

-- Display all records in Members table (John Carter should be removed)
SELECT * FROM Members;

-- Display all records in Loans table
SELECT * FROM Loans;

```

Source Code:

https://github.com/mubashir1837/CS-FA-25/blob/main/CS%202203-01%20-%20AY2026-T4/APRIL-30/library_management_system.sql

Conclusion

This assignment demonstrates the practical application of SQL in designing and managing a relational database. By carefully selecting data types , VARCHAR for text fields, INT for numeric identifiers, DATE for temporal data , the schema ensures data integrity and storage efficiency. DDL commands were used to create the normalized three,table schema, while DML commands (INSERT, SELECT, UPDATE, DELETE) performed all required data operations.

The use of primary keys, foreign keys, and NOT NULL constraints enforces referential integrity across the Loans relationship. The SELECT,JOIN query illustrates how relational data spread across multiple tables can be retrieved as a unified, meaningful result. As Peterson (2023) notes, mastering these fundamental SQL commands is essential for effective database management in any real,world application.

References

CBT Nuggets. (2019, July 14). *What is the SQL language?* [Video]. YouTube.

Peterson, R. (2023, December 16). SQL cheat sheet with commands & description (2024). *GURU99.* https://www.guru99.com/sql,cheat,sheet.html

SQL: How to create and delete a table with SQL query. (n.d.). *EnableGeek.* https://www.enablegeek.com/tutorial/create,delete,table,sql,query/

Vidhya, V., Jeyaram, G., & Ishwarya, K. (2016). *Database management systems.* Alpha Science International.